

Curriculum Plan

Introduction to Engineering with Microelectronics

Time

5 Class Sessions – Approx. 10 hours

Description

Construct various electronic circuits on a breadboard. These circuits will be controlled by the Arduino UNO microprocessor.

Prerequisite Knowledge

Learners:

- Follow instructions
- Basic computer skills
- Understanding of basic electronic components such as pushbuttons and switches.
- Reading simple wiring diagrams
- Using multimeters for voltage or continuity checks

Instructors:

- Working knowledge of Arduino. Power requirement, how to load programs, etc.
- Arduino software proficiency
- Breadboard and circuit expertise
- Basic programming concepts
- Electrical safety and lab management
- Explain concepts using multiple instructional strategies such as visual, verbal and hands-on

Learning Outcomes/Goals

- Introductory knowledge of solid-state devices, such as, diodes, LED's and transistors.
- Introductory knowledge of PWM Pulse Width Modulation, sensors, potentiometers, photoresistors, etc.
- Basic understanding of microcontroller programming language.

Summative Assessments

Student Evidence Requirements

- Photos / screenshots of setup
- Data table completed (expected vs actual)
- Code / configuration file attached (if applicable)
- Short written analysis tied to SCALE competencies
- Reflection on lifecycle implications (manufacture/deploy/sustain)

Instructor Assessment

See Grading Rubric in “Presentations and Supplemental Materials” at the end of this document.

Resources

What resources will instructors and learners need to complete the lessons?

- Computer – Provided by school
- UNO R3 Project Kit – Provided by school
- Super Starter Kit Uno R3 Project Tutorial (ELEGOO, n.d.) – Provided by school
- Notebook – Provided by student

Lesson 1: Arduino Serial Communication and Basic Circuit Control

Time

2 Hours

Learning Objectives

- Familiarization with the Arduino microprocessor and the software used to program it.
- Learn how to construct simple circuits on a breadboard

This lesson aligns with the following KSAs:

- Wiring and testing a control circuit
- Using diagnostic tools (multimeter, oscilloscope)
- Independently identify a fault and justify the corrective action
- Processes, workflows, and system architecture

Introduction

In this lesson you will make a connection to the Arduino with your computer and setup the Serial Monitor. Once that connection is established you will connect three different circuits and load a different program for each.

Resources

Leaners:

- Arduino UNO R3 Project Kit (Provided by school)
- Desktop or Laptop Computer (Provided by school)
- Arduino Integrated Development Environment Software (Provided by school)
- Notebook (Provided by student)

Classroom Instruction

Instructor Activities	Learner Activities	Assessment
Arduino Basics Presentation	Take notes	n/a
Assist students	Setup communication w/ Arduino	Serial Monitor working
Assist students	Complete “Blink” lab	Working circuit

Summary

In this lesson students will complete the first three lessons in the Arduino Lab Manual.

Skills covered:

- Loading a program from the library
- Setting up and viewing the serial monitor
- Connecting simple circuits on the project breadboard
- Basic understanding of a simple solid state device.

Assignment

Complete each lab as instructed and demonstrate to instructor

Additional Instructor Notes

Assessment Questions:

- What is the first line of code in the Blink program?
- How is the polarity of an LED marked?
- How is the value of a resistor displayed?
- Why is it important to put a resistor in series with an LED?
What is the code to change the blink speed of the LED?

Additional Learner Notes

- Always save a backup of your program.
- Take pictures of your circuit for reference and credit.

Appendix/Attachments/References

ELEGOO. (n.d.). *Super starter kit UNO R3 project tutorial*. <https://www.elegoo.com>

Arduino IDE Software - <http://www.elegoo.com/download/>.

Lesson 2: Digital Electronics

Time

4 Hours

Learning Objectives [include alignment to learning outcomes and KSAs]

- **Explain** how RGB LEDs produce multiple colors using additive color mixing.
- **Identify** the function of current-limiting resistors in LED circuits.
- **Wire** an RGB LED correctly to an Arduino Uno using a breadboard.
- **Write and upload** an Arduino sketch to control RGB LED colors using PWM outputs.
- **Modify** code parameters to produce specific colors and brightness levels.

This lesson aligns with the following KSAs:

- Basic digital electronics concepts (digital inputs vs. digital outputs)
- Additive color mixing principles used in RGB LEDs
- Electrical characteristics of LEDs and the need for current-limiting resistors
- Properly wire an RGB LED and pushbutton using a breadboard and Arduino Uno
- Select and install appropriate current-limiting resistors
- Configure Arduino digital input and output pins
- Integrate hardware and software components into a functioning digital system
- Modify and adapt code to achieve specific functional outcomes
- Troubleshoot wiring and programming issues independently

Introduction

In this lab, you will explore how an Arduino Uno R3 interacts with common electronic input and output devices. Using components from the Arduino R3 Starter Kit, you will learn how microcontrollers process digital inputs and control visual and audible outputs such as an RGB LED and active and passive buzzers. These devices are widely used in automated systems to provide user feedback, alerts, and status indicators.

Resources

ELEGOO. (n.d.). *Super starter kit UNO R3 project tutorial*. <https://www.elegoo.com>

Arduino IDE Software - <http://www.elegoo.com/download/>.

ELEGOO Super Starter Kit Uno R3 Project (or latest version)

Learners:

- Arduino Language Reference
- RGB LED Color Mixing Charts
- Basic Electronics Reference Sheet
- Sample Arduino Sketches

Instructors:

- Wiring Diagrams
- Instructor Demo Sketches
- Troubleshooting Guide
- Assessment Tools

Classroom Instruction

Instructor Activities	Learner Activities	Assessment
Embedded Systems Presentation	Complete Lessons 4-7 in the Arduino manual	Grading Rubric
Electronic Sensors Presentation	Demonstrate each circuit to instructor	
Grade completed circuits		

Summary

This lab emphasizes not only correct wiring and coding practices, but also the ability to modify, test, and troubleshoot a system to achieve specific performance goals. The skills developed in this activity form a foundation for more advanced topics in automation, robotics, and embedded control systems, and align with industry expectations for entry-level technicians and technologists.

Assignment

- Correctly wiring the required Arduino circuits (RGB LED, digital input, and buzzer)
- Writing and uploading functional Arduino sketches for each device
- Demonstrating system operation to the instructor or submitting required evidence (photos, code files, and reflection)
- Completing a brief post-lab reflection explaining how inputs affect outputs and how code modifications changed system behavior

Additional Instructor Notes

- Why do I need a resistor on the RGB LED?
(Overcurrent protection; different colors have different forward voltages.)
- Why isn't my RGB LED showing the correct color?
(Incorrect pin order, common anode vs. common cathode, PWM pin usage.)
- Why does my button input behave inconsistently?
(Floating inputs; need for pull-up or pull-down resistors or INPUT_PULLUP.)
- Why does the active buzzer only turn on/off while the passive buzzer plays tones?
(Internal oscillator vs. externally driven frequency.)
- Why does my code compile but not behave correctly?
(Logic errors, incorrect pin assignments, missing pinMode() statements.)

Additional Learner Notes

- Students new to breadboarding may require additional setup time.
- Debugging wiring issues often takes longer than coding—plan buffer time.
- Consider breaking the lab into checkpoints (one device at a time).
- Faster students can be extended with optional challenges (custom colors, melodies, or input-controlled behavior).

Appendix/Attachments

ELEGOO. (n.d.). *Super starter kit UNO R3 project tutorial*. <https://www.elegoo.com>
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Lesson 3: Pulse-Width Modulation

Time

4 Hours

Learning Objectives

- Explain how pulse-width modulation (PWM) controls servo motor position.
- Identify power and signal requirements for servo motor operation.
- Connect a relay module to an Arduino digital output.
- Develop an Arduino sketch to control relay switching.
- Apply sensor data to influence system behavior.
- Integrate joystick input to control system outputs.

This lesson aligns with the following KSAs:

- Motion Control
- Power and Safety
- Sensor Feedback
- Human Input
- Remote Control

Introduction

In this lab sequence, you will expand your Arduino Uno R3 skills by integrating actuators, sensors, and user input devices into a functional control system. Building on earlier lessons focused on basic inputs and outputs, you will now work with more advanced components including a servo motor, relay module, environmental sensor (temperature or light), joystick module, and an infrared (IR) receiver with remote control.

Resources

ELEGOO. (n.d.). *Super starter kit UNO R3 project tutorial*. <https://www.elegoo.com>
Arduino IDE Software - <http://www.elegoo.com/download/>.

Classroom Instruction

Use the table below to explain the instructor activities/instruction, learner activities, and assessments in detail. If an instructor decides to use this lesson, this is the information they will need to facilitate it. For assessments and

handouts, use detailed naming conventions.

Instructor Activities	Learner Activities	Assessment
Confirm Arduino IDE installation and required libraries (e.g., IRremote).	Construct circuits for each component using a breadboard.	Instructor check-ins and questioning
Prepare example wiring diagrams and starter sketches.	Modify provided code to meet defined performance requirements.	Observation of troubleshooting behavior
Explain how each lesson builds toward a complete automation system.	Troubleshoot wiring or logic errors independently.	Submit required documentation.

Summary

This lab sequence requires you to not only follow procedures, but also to modify, integrate, and troubleshoot multiple components as part of a complete system. By the end of the lab, you will be able to design and implement a simple Arduino-based automation solution that reflects real-world applications found in manufacturing, robotics, and industrial control systems. The knowledge, skills, and abilities developed here directly support the course learning outcome and prepare you for more advanced automation and embedded systems work.

Assignment

To successfully complete the assignment, learners must:

- Correctly wire all required components
- Develop and upload functional Arduino code
- Demonstrate system operation and responsiveness
- Explain how inputs are processed and how outputs are controlled
- Troubleshoot and resolve basic wiring or logic errors

Additional Instructor Notes

- Expect longer setup time for servo and relay wiring than previous lessons.
- IR receiver configuration and library installation may require additional troubleshooting time.

- Consider planning for 2–3 lab sessions or provide clear checkpoints if delivered in a single extended session.
- Build in buffer time for debugging and system integration.

Additional Learner Notes

- Build and test one component at a time. Verify each device works before combining them into a larger system.
- Double-check wiring before powering the circuit. Many issues are caused by misplaced wires or missing ground connections.
- Label or comment your code clearly. This will help you troubleshoot and explain your work later.
- Save working versions of your sketches. If something stops working, you can return to a known-good version.
- Use the Serial Monitor. Printing values and messages is one of the most effective debugging tools.
- Be patient with troubleshooting. Debugging is an expected and valuable part of learning automation systems.

Appendix/Attachments

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Presentations and Supplemental Material

Arduino Basics Presentation (Free download) - Review/Discuss Lessons 0 and 1 in the Elegoo Tutorial ELEGOO. (n.d.)

Embedded Systems Presentations

What is an Embedded System https://youtu.be/oPn_adlC1Q0?si=D4yGtxt4HmR2pZXI

So, You Want to be an Embedded Systems Engineer -

<https://youtu.be/m8w2FzqU5jg?si=fuvEe8QXWVznFUl>

Electronic Sensors Presentation

https://youtu.be/Xl49uFm5HRE?si=_Kh0s2yHHfQOYq3l

Other resources for educational material on the web

All About Circuits

<https://www.allaboutcircuits.com/>

Engineering Mindset

<https://www.youtube.com/@EngineeringMindset>

ElectroBOOM

<https://www.youtube.com/@ElectroBOOM>

Grading Rubric (Instructor Use)

Category	Criteria	Points	Score
Safety & Professional Practice	Follows safety rules, uses tools correctly, maintains clean station	10	
Procedure Execution	Completes required steps; demonstrates proper setup and process control	25	
Data Quality	Accurate measurements/logs; appropriate units; repeats/controls noted	20	
Analysis & SCALE Relevance	Explains results; identifies risks/vulnerabilities; lifecycle considerations	25	
Documentation	Clear, complete, well-organized submission; evidence attached	20	

Total Score: _____ / 100

Feedback / Notes: